Linear carbon-cycle impulse responses: post-cessation monotonicity, certified peak delays, and two concentration maxima from a single emission hump

(a model result; the unimodality theory is classical)

Xavier Callens (SocrateAI Lab)

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Abstract

We study the *linear* impulse-response (IRF) model $u(t) = \int_0^t m(z)\, h(t-z)\, dz$ with the Joos et al. (2013) multi-model-mean IRF $h(t) = a_0 + \sum_{i=1}^{3} a_i e^{-t/\tau_i}$ [1], for emission paths $m \geq 0$. "Concentration" always means the model excess u; nothing is said about the real atmosphere. **(a)** After emissions cease, u is completely monotone, hence non-increasing and convex, with limit $a_0 \int m$. Hand proof for continuous m (Tier [C]); Lean proof for *discrete nonnegative emission measures only* (Tier [A], statement lock owed). **(b)** For Seshadri's emission family [2], interval-arithmetic enclosures of the time of the concentration maximum and of its delay after the emissions peak, in 63 scenarios (Tier [C]); delays of 50–227 yr in the Joos table. **(c)** For every completely monotone, strictly decreasing IRF that is not a single exponential (in particular the Joos IRF), there are single-humped emission paths for which u has two local maxima separated by a strictly lower value; an explicit staircase family with an exactly characterised window $B/A \in (h(d), L)$ is proved (Tier [C]), and three Joos instances are certified with exact rationals and interval arithmetic, one of them for a strictly unimodal emission path. **The mechanism is classical**: it follows from Ibragimov's 1956 theorem on strong unimodality (convolution preserves unimodality iff the kernel is log-concave), which we read only from secondary sources; the single exponential is its log-concave special case. What is added is the explicit window, the certified Joos instances and the carbon-cycle framing. To our knowledge (not a systematic search) no carbon-cycle paper states it. No tonnes, climate effect, policy or "discovery" claim is made.

What this is NOT. It is **not** a statement about the real carbon cycle, which is nonlinear and state dependent; **not** about realistic emission pathways (the inputs giving two maxima are mathematical test inputs); **not** about warming, tonnes or policy; **not** a new phenomenon class; and the Lean part covers only discrete emission measures (T2, T3, part (b) and the continuous case are not in Lean).

Markers. [A]: kernel-checked Lean, axioms within {`propext`, `Classical.choice`, `Quot.sound`}, controls caught, no statement lock yet. [L]: a cited source, taken as stated. [C]: executed code or a hand proof with the argument written down, not machine-checked. [X]: floating-point scans and comparisons, supporting nothing by themselves. Identifiers `CK3-·-nnnn` refer to `ledger/ledger.json`.

1 Model, definitions and scope

Model (Tier [L]). Following Seshadri [2] (eq. 2) the concentration is $u(t) = u_P + \int_0^t m(z)\, h(t-z)\, dz$, where m is the emission rate in concentration units (the rate at which u would rise if the lifetime were infinite) and the IRF h, with $h(0) = 1$, is the fraction of an added pulse still airborne after time t, as defined by Joos et al. [1] (eq. 7 and text). We work with the excess $u - u_P$, written u. A *concentration maximum* is a local maximum of this model excess u. The Joos IRF is $h(t) = a_0 + \sum_{i=1}^{3} a_i e^{-t/\tau_i}$ with (Table 5, row IRFCO2) $a_0..a_3 = 0.2173, 0.2240, 0.2824, 0.2763$ and $\tau_1..\tau_3 = 394.4, 36.54, 4.304$ yr; the a's sum to 1 exactly (control P1). They are used as exact decimals. The Joos paper is distributed under CC BY 3.0 (journal landing page and PDF header).

Unimodal and "two maxima". An emission path $m \geq 0$ is *weakly unimodal* if it is non-decreasing and then non-increasing, and *strictly unimodal* if it has a unique mode and is strictly monotone on each side within its support. "u has two maxima" means: two local maxima of u *separated by a strictly lower value*. The certificates below, $u(s_1) < u(s_2) > u(s_3) < u(s_4) > u(s_5)$ at times $s_1 < \cdots < s_5$, establish exactly this (continuity then gives a local maximum in (s_1, s_3) and one in (s_3, s_5), with $u(s_3)$ strictly below both).

Fit window. Joos et al. state that their fit (eq. 11) applies for $0 \leq t \leq 1000$ yr only. The limit $a_0 \int m$, the check $u(10^6)$ and the scan of u' on $[0, 6000]$ yr below are properties of the *fitted formula* beyond its stated range, not of the multi-model response. The certified peak times of part (b) (≤ 287 yr) and the certified times of part (c) (≤ 500 yr) lie inside the window.

Scope. Linear model, multi-model-mean coefficients for one pulse experiment. No tonnes, no climate effect, no emission-pathway or deployment advice. "Concentrations fall after emissions stop" is a model statement.

2 Prior art and what is added

By Ibragimov [3] (1956, "On the composition of unimodal distributions"; "strong unimodality"), convolution with every unimodal density preserves unimodality *if and only if* the kernel is log-concave. We read this only from secondary sources (the Encyclopedia of Mathematics entry "Unimodal distribution" and Watanabe, Nagoya Math. J. 1991 [4]), not from the original (`CK3-L-0003`, Tier [L]). Consequences:

- The one-sided exponential $e^{-t/\tau}\mathbf{1}[t \geq 0]$ is log-concave, so **Theorem 2 below is a special case of Ibragimov's theorem.**
- A completely monotone (CM) kernel that is not a single exponential is strictly log-convex on $(0, \infty)$, so Ibragimov's converse already implies, *for integrable kernels* ($a_0 = 0$), that some unimodal input yields a non-unimodal output. This is the existence half of part (c) for $a_0 = 0$.
- For $a_0 > 0$ the IRF is not integrable and the theorem as stated does not apply directly; Theorem 3 gives an explicit construction covering both cases.

Added here: the explicit staircase window $B/A \in (h(d), L)$, its necessity within the family, the certified Joos instances (including a strictly unimodal path), the certified peak delays of part (b), and the carbon-cycle framing. To our knowledge (Joos 2013, Seshadri 2016, one alphaXiv query and two web searches; *not* a systematic search; the literature on delays to peak *warming*, e.g. MacDougall–Ricke–Caldeira 2014 and Zickfeld–Herrington 2015, was not read for this question) the two-maximum behaviour is not stated in the carbon-cycle literature read. The physical intuition is elementary: with $a_0 > 0$, any sustained positive emission makes the model u grow, so the end of a long low plateau of emissions produces a second maximum (a model property only). The result is an explicit, certified instance of a known phenomenon class, not a new phenomenon.

3 Results

Throughout, $h(t) = a_0 + \sum_i a_i e^{-t/\tau_i}$, $a_0 \geq 0$, $a_i > 0$, $h(0) = 1$. Then $h(x) = \int e^{-\lambda x} \mu(d\lambda)$ for a probability measure μ with atoms at 0 (mass a_0) and $1/\tau_i$ (mass a_i).

3.1 (a) Post-cessation complete monotonicity

Theorem 1 (T1; Tier [C] for continuous m, Tier [A] for discrete emission measures)**.** *Let $m \geq 0$ be supported on $[0,T]$, $W = \int m$. For $t > T$, $u(t) = a_0 W + \sum_i b_i e^{-t/\tau_i}$ with $b_i = a_i \int e^{z/\tau_i} m(z)\,dz \geq 0$. Hence $(-1)^n u^{(n)}(t) = \sum_i b_i \tau_i^{-n} e^{-t/\tau_i} \geq 0$ for all n: u is completely monotone on (T,∞), in particular non-increasing and convex, with limit $a_0 W$ (the limit is a property of the fitted formula beyond 1000 yr).*

This is elementary (a positive mixture of shifted CM functions) and, to our knowledge, standard; its value here is an independent kernel check, not novelty.

Negative control. With one negative coefficient ($a_0 = 0.6573$, $a_1 = -0.2240$, $a_2 = 0.2824$, $a_3 = 0.2763$, sum 1) 181 of 200 random histories violate monotonicity or convexity, and $h = 1 - e^{-t}$ is not CM in Lean (`neg_ctl_not_cm`).

3.2 (c) Single-humped emissions and the number of concentration maxima

Theorem 2 (T2, single exponential; Tier [C]; special case of Ibragimov)**.** *Let $h = e^{-t/\tau}$ and let $m \geq 0$ be weakly unimodal (of locally bounded variation). Then u is non-decreasing and then non-increasing (quasi-concave). Its maximisers form an interval, and u never has two local maxima separated by a strictly lower value.*

Proof. $u(t) = \int_0^t m(z) e^{-(t-z)/\tau} dz$ gives $u' = m - u/\tau$. Put $V(t) = \int_0^t e^{z/\tau} m(z)\,dz = e^{t/\tau} u(t)$ and $g = e^{t/\tau} u' = e^{t/\tau} m - V/\tau$. Then $dg = e^{t/\tau} dm$, so g is non-decreasing up to the emission peak, with $g(0) = m(0) \geq 0$, and non-increasing after it. Hence g, and so u', is ≥ 0 and then ≤ 0. □

Remark on wording. The statement "u has at most one local maximum" is literally false for non-strict maxima: an impulse $c\tau$ at time 0 (allowed in the discrete setting) followed by constant emission c gives $u \equiv c\tau$, and every point of the plateau is a local maximum. Only the quasi-concave formulation above is claimed. No counterexample to it was found in 400 adversarial random staircases (auditor, Tier [X]).

Theorem 3 (T3; Tier [C])**.** *Let h be completely monotone, $h(0) = 1$, strictly decreasing and not a single exponential. Let $d > 0$ and $\rho(x) = h(x+d)/h(x)$. Then ρ is strictly increasing, with $\rho(0) = h(d) < \rho(\infty) = L \leq 1$, where $L = 1$ iff $a_0 > 0$. For $A > 0$ and $\theta = B/A \in (0,1)$ let*

$$m = A \text{ on } [0,d), \qquad m = A - B \text{ on } [d, z_3), \qquad m = 0 \text{ afterwards},$$

a weakly unimodal (non-increasing) nonnegative path, with $z_3 > d + x^*$ where $\rho(x^*) = \theta$ whenever $\theta \in (h(d), L)$. If $\theta \in (h(d), L)$, then u has exactly two *local maxima, at $t = d$ and $t = z_3$, and one local minimum, at $t = d + x^*$. If $\theta \leq h(d)$ or $\theta \geq L$ it has exactly one. So the window $B/A \in (h(d), L)$ is necessary and sufficient within this family.*

Proof. (i) *Strict monotonicity of ρ.* By Bernstein, $h = \int e^{-\lambda x} \mu(d\lambda)$; Cauchy–Schwarz gives $hh'' \geq h'^2$, with equality only if μ is a point mass, i.e. h is a single exponential (this includes the constant $h \equiv 1$, excluded by strict decrease). So $\log h$ is strictly convex and $\log \rho(x) = \log h(x+d) - \log h(x)$ is strictly increasing. $\rho(0) = h(d)$; $\rho(\infty) = 1$ if $a_0 > 0$ and $e^{-d/\tau_{\max}}$ if $a_0 = 0$.
(ii) *Sign of u'.* For a staircase with jumps c_j at z_j, $u'(t) = \sum_j c_j h(t - z_j)$ off the jump times. Here the jumps are $+A$ at 0, $-B$ at d, $-(A-B)$ at z_3. On $(0,d)$, $u' = Ah > 0$. For $t = d + x \in (d, z_3)$, $u'(t) = Ah(x+d) - Bh(x) = h(x)\big(A\rho(x) - B\big)$, which is negative for $x < x^*$ and positive for $x > x^*$. Further $u'(d-) = Ah(d) > 0$ and $u'(d+) = A(h(d) - \theta) < 0$: a local maximum at d, and by the sign pattern a local minimum at $d + x^*$.
(iii) *At z_3.* $u'(z_3-) = Ah(z_3) - Bh(z_3 - d) > 0$ since $z_3 - d > x^*$, and

$$u'(z_3+) = -A\big(1 - h(z_3)\big) + B\big(1 - h(z_3 - d)\big) < 0,$$

because $A > B$ and $1 - h(z_3) > 1 - h(z_3 - d) \geq 0$ (h strictly decreasing, $h \leq 1$). So z_3 is a local maximum. After z_3, u is CM by Theorem 1, hence decreasing: no further maximum. Together with $u' > 0$ on $(0,d)$ there are exactly two maxima.
(iv) *Necessity.* If $\theta \leq h(d)$ then $\rho(x) > h(d) \geq \theta$ for $x > 0$, so $u' > 0$ on (d, z_3), u increases on $(0, z_3)$ and the only maximum is at z_3. If $\theta \geq L$ then $\rho < L \leq \theta$, $u' < 0$ on (d, z_3) and the only maximum is at d. □

Corollary 4 (headline of (c); Tier [C]). *For every IRF h with $h(0) = 1$ that is completely monotone, strictly decreasing and not a single exponential (in particular the Joos IRF), there is a weakly unimodal emission path for which u has two local maxima separated by a strictly lower value. For a single exponential, u is quasi-concave for every weakly unimodal path (Theorem 2).*

Theorem 3 uses a *weakly* unimodal path (a non-increasing step function, mode at 0). That the existence also holds under the *strict* definition is shown for the Joos IRF by the certified instance I2 below, not by a general proof.

Mechanism (Tier [X], model property only). With $a_0 > 0$, any sustained positive emission makes the model u grow once the fast modes ($\tau = 4.3$ and 36.5 yr) have decayed, so the second maximum is the end of the low plateau. This is expected once stated.

3.3 Certified instances (Tier [C])

Five times $s_1 < \cdots < s_5$ per instance, exact rational inputs from Joos Table 5, mpmath interval arithmetic at 60 digits (`data/data/results_c.json`; half-widths $< 10^{-56}$); the certificate is $u(s_2) > u(s_1)$, $u(s_2) > u(s_3)$, $u(s_4) > u(s_3)$, $u(s_4) > u(s_5)$.

instance and emission path	s (yr)	$u(s_1), \ldots, u(s_5)$	lower bounds of the four differences
I1: continuous piecewise linear, *weakly* unimodal: nodes (yr, level) $(0,0), (10,1), (22, \frac{1}{10}), (300, \frac{1}{10}), (310, 0)$; non-decreasing to year 10, then non-increasing with a plateau on $[22, 300]$	10, 21, 32, 303, 320	4.2161, 7.9964, 7.6834, 15.4774, 14.9042	3.78, 0.313, 7.79, 0.573
I2: nodes $(0,0), (10,1), (22, \frac{1}{10}), (300, \frac{1}{20}), (310, 0)$; strictly increasing on $[0, 10]$, strictly decreasing on $[10, 310]$, zero afterwards; continuous. **Meets the strict definition.**	10, 21, 40, 305, 330	4.2161, 7.9964, 7.7188, 12.1543, 11.5925	3.78, 0.278, 4.44, 0.562
S1: staircase, jumps $+1$ at 0, $-\frac{9}{10}$ at 5 yr, $-\frac{1}{10}$ at 200 yr (a non-increasing step function, weakly unimodal, mode at 0)	1, 5, 11, 200, 500	0.9661, 4.3361, 4.0146, 10.6010, 7.2520	3.37, 0.322, 6.59, 3.35

The float picture of I1 has maxima at 20.7 and 303.2 yr and a minimum at 31.9 yr. A first attempt for S1 with the wrong middle time ($s_3 = 100$) returned `two_local_maxima_certified: false`; the times were then chosen from the float minimum (11 yr). The certificate depends on the chosen times, not on a search that hid failures. The instances were recomputed by the auditor with independent code (Section 7).

3.4 Where it happens (Tier [X], floats)

Plateau family (rise to 1 by yr 10, linear drop to level L over D years, plateau to yr 300, ramp to 0 by yr 310): two maxima for $L \le 0.15$ for all $D \in \{2, 5, 12, 20, 40\}$; for $L = 0.2$ for $D = 2, 5, 40$ but one maximum for $D = 12, 20$; one maximum for $L \ge 0.25$. Over 1500 random continuous unimodal piecewise-linear paths (seed 7) with the Joos IRF: 17 have no interior maximum, 1364 one, 117 two, 2 three (the three-maximum cases are float-grid counts and are not certified; the audit did not check them); the same paths with a single exponential never give more than one (control P2). A hypothetical IRF with $a_0 = 0$ (decaying modes only) also gives 2 maxima with a long plateau; a two-term IRF $a_0 = a_1 = \frac{1}{2}$, $\tau = 50$, gives 2 maxima for $B/A \in \{\frac{19}{20}, \frac{49}{50}\}$ and 1 for $\frac{4}{5} < h(d)$, as Theorem 3 predicts (`logs/check_two_term.log`). A sufficient condition for "at most one peak" beyond the single exponential was not found.

4 (b) Peak times and delays for Seshadri's emission family

Family. $m(t) = m_0(1+r)^{\min(t,t_g)-t_0} e^{-(t-t_0)/\tau_m}$ [2]. Our setup puts the origin at $t_0 = 0$ and ignores emissions before t_0 (not varied). Seshadri integrates from an earlier origin $0 < t_0$ (his m'_{av0} term), so the producer's time origin differs from his and agreement with his figures is Tier [X] only.

Method (Tier [C]). Closed-form u' and u'' in interval arithmetic (mpmath `iv`, 60 digits), exact rational bisection with certified signs $u'(\text{lo}) > 0 > u'(\text{hi})$ and $u'' < 0$ on the initial bracket (width 0.01 yr): the peak is the unique local maximum *inside the bracket*, with enclosure width $< 10^{-19}$ yr. The emissions peak t_1 is certified from the sign of $\ln(1+r) - 1/\tau_m$: $t_1 = t_g$ if positive, $t_1 = t_0 = 0$ if negative. 63 scenarios (3 IRF variants $\times$ 3 values of $r \times 7$ values of τ_m) are all CERTIFIED (`data/data/results_b.json`). **Global uniqueness of each peak is Tier [X]**: a float scan of u' on $[0, 6000]$ yr found exactly one $+ \to -$ sign change in all 63 scenarios; the range exceeds the 1000-yr window of the fit, so this describes the fitted formula, not the multi-model response.

Delays $t_2 - t_1$ in years for Joos Table 5, $t_g = 60$ yr (two decimals; enclosures in the JSON):

τ_m	$r = 0.03$	$r = 0.05$	$r = 0.015$
20	69.55 ($t_1 = 0$)	79.34 ($t_1 = 0$)†	58.68 ($t_1 = 0$)
40	50.47	59.52	99.01 ($t_1 = 0$)
50	70.48	79.40	119.04 ($t_1 = 0$)
60	91.22	100.12	139.78 ($t_1 = 0$)
80	134.32	143.31	122.80
100	176.95	185.99	165.36
120	217.35	226.37	205.73

† Convention-dependent: $\ln(1.05) = 0.0488 < 1/\tau_m = 0.05$ gives $t_1 = 0$ in our exact $(1+r)^t$ reading. Under Seshadri's e^{rt} approximation $r = 1/\tau_m$ exactly, emissions are flat on $[0, 60]$ and t_1 is undefined.

- The delay is **not monotone in** τ_m across the boundary $r = 1/\tau_m$, where t_1 jumps from t_g to 0; the peak time t_2 itself is continuous there (the auditor finds 98.28871 on both sides at $\ln(1.03) = 1/\tau_m$, $\tau_m \approx 33.83086$, and the delay jumps by exactly $t_g = 60$ yr). The table is not a smooth trend.
- *Seshadri comparison, Tier [X].* The parameters of his Fig. 5 ($t_g - t_0 = 60$ yr, $r = 0.015$ %/year, $\tau_m = 50$ yr) fall outside the case $r > 1/\tau_m$ to which his Sect. 3.2 restricts attention, under either reading of "0.015 %/year" (0.015/yr or 0.00015/yr; both are below $1/\tau_m = 0.02$, and $\ln 1.015 = 0.0149$). This is not resolved here. For $r = 0.015$, $\tau_m = 50$ our certified value is 119.04 yr (with $t_1 = 0$), inside the 100–135 yr he states for that figure, and his Fig. 2 range of 40–220 yr is reproduced in order of magnitude. This agreement does not show the intended reading of his parameters. Separately, his printed eq. (4) omits the boundary term $\sum_i \mu_i e^{-t/\tau_i} m(0)$ from the integration by parts and is exact only when $m(0) = 0$ (stated neutrally; the author was not contacted).
- His printed eq. (1) has coefficients $0.276, 0.282, 0.224, 0.217$ with sum 0.999 (a neutral factual note), and the text uses $\tau_1 = 4.4$. Variants with these coefficients and $\tau_1 = 4.30$ or 4.4 change the delay by at most 0.36 yr across all 21 matched scenario pairs, so the 0.001 coefficient defect is immaterial for the delays, though it matters for the long-run limit.

Positive control. For $h = e^{-t/30}$, $m = e^{-t/50}$ the closed-form peak is $75\ln(5/3) = 38.311921782449301240\ldots$; the certified enclosure [38.31192178244930124039, 38.31192178244930124045] contains it (`data/data/control_closed_form.json`).

5 Controls

control	result
P1 coefficient sums	Joos sums to 1; Seshadri's printed eq. (1) to 0.999 (exact rationals); pass
P2 single exponential	closed-form u vs 1500 random unimodal paths: at most 1 maximum; pass
P3 closed form vs quadrature	closed-form u equals mpmath quadrature of the convolution in 50/50 points, tolerance 10^{-20}; pass
P4 arithmetic (Ciais et al. [5], printed numbers only)	$10.38 - 2.04 - 3.15 - 2.06 \cdot 2.124 = 5091/6250 = 0.81456$, rounds to the printed 0.81; pass
P5 post-cessation	200/200 random histories antitone and convex on a grid; $u(10^6) = a_0 W$ to about 10^{-8} in 5 spot checks (beyond the fit window); pass
Closed-form peak	$75\ln(5/3)$ enclosed; pass
N1 negative coefficient	181/200 histories violate; Lean `neg_ctl_not_cm`; pass
N2 wrong sign in H $(1 + e^{\cdot})$	rejected at 50/50 points; pass
N3 single exponential vs Joos, same emission	1 vs 2 maxima (I1 and S1); pass
N4 planted `sorry`/axiom in `Controls.lean`	compiles, rc 0, caught by `#print axioms`; pass

6 Lean (Tier [A], discrete nonnegative emission measures only)

Module `formal_cert/Programmes/CarbonK3/Core.lean` (copy in `data/lean/`) imports `Programmes.CompletelyMonotone.Basic` (the repository's CM library; never `import Mathlib`), compiled with `scripts/certified_numerics/lean_pinned.py` (Lean 4.34.1, Mathlib `d13f23b7`), rc 0 (17.7 s; the first run took 22 min on a cold disk cache and failed on API names). The definitions are $\mathrm{irf}(t) = a_0 + \sum_i a_i e^{-r_i t}$ and $\mathrm{excess}(t) = \sum_j w_j \,\mathrm{irf}(t - z_j)$ (a finite sum of impulses $w_j \geq 0$ at times z_j). Theorems: `irf_cm` (for $a_0, a_i, r_i \geq 0$ the IRF is CM), `excess_cm_after` (for $w_j \geq 0$, $z_j \leq T$, $s \mapsto \mathrm{excess}(s + T)$ is CM on $(0, \infty)$), `excess_antitone`, `excess_convexOn`, `excess_tendsto` (limit $a_0 \sum_j w_j$ when all $r_i > 0$; no sign hypotheses), `neg_ctl_not_cm` ($1 - e^{-t}$ is not CM); the helper `cm_convexOn` was added during the run. `#print axioms` of all six: {`propext`, `Classical.choice`, `Quot.sound`}. `leanchecker Programmes.CarbonK3.Core`: rc 0. Escape grep (`sorry`, `native_decide`, `admit`, `unsafe`, `extern`, `implemented_by`, `axiom`, `opaque`, `set_option`, `csimp`) on `Core.lean`: no match.

Controls. `Controls.lean` compiles with rc 0 and is caught by the axiom gate: `ctl_sorry` shows `sorryAx`, `ctl_axiom` shows `CarbonK3.ctl_cheat`.

Scope and caveats. `IsCompletelyMonotoneOn` (`Basic.lean` l.28) is the genuine definition: C^∞ on $(0, \infty)$ and $(-1)^n f^{(n)} \geq 0$. `excess` is defined for all t without a Heaviside factor; it equals the model excess only for $t \geq \max_j z_j$, which is the only range the theorems use. The auditor recompiled `Core.lean` (rc 0, 13.6 s), ran `leanchecker` (rc 0, 61.4 s) and obtained identical axiom lists; the dependency oleans (`Programmes.CompletelyMonotone.*`, `Autoresearch.KeldyshCM`) were *reused, not rebuilt* (axiom audit and leanchecker are transitive, so they still cover them). **Not in Lean:** continuous emission paths, Theorems 2 and 3, part (b). **Statement lock owed**; no person has certified that the statements mean what their gloss says.

- `Core.lean`: `00e6ba6c08f4fffcb9b9162eb7921b0a756475a55c4cb8d722ea364aa49e363a`
- `Controls.lean`: `a56d76fb410ce843dbadfced5094b825cee5a523b1ab022849b461673d71c3ff`

7 Independent AI audit

A different AI agent instance (`AUDIT.md`, 2026-10-31; verdict ACCEPT_WITH_CORRECTIONS; none of the producer's code used; its scripts are in `data/audit_scripts/`, small outputs only, oleans not shipped) recomputed:

- *Sources:* all nine Joos Table 5 numbers, the $0 \le t \le 1000$ yr validity of eq. (11), and Seshadri's eqs. (1), (2) and Sect. 3.2.
- *(c) instances:* Sage `RealBallField` at 400 bits with its own closed-form segment integrals; the u values at all 15 times agree with `results_c.json` to the 20 printed digits (ball radii $\le 5 \cdot 10^{-116}$); lower bounds of the four differences I1: 3.780, 0.3130, 7.794, 0.5732; I2: 3.780, 0.2776, 4.435, 0.5618; S1: 3.370, 0.3216, 6.586, 3.349.
- *(b):* own closed form with mpmath at 50 digits, step 0.5 yr on $[0, 3000]$, then `findroot`: for all 21 Joos scenarios exactly one $+ \to -$ change of u' (a different grid and end point from the producer's, hence a Tier [X] cross-check, not an independent interval certificate of uniqueness); t_2 agrees with the enclosures to $\le 6.0 \cdot 10^{-20}$, t_1 in every cell; Seshadri-coefficient variants for three (r, τ_m) cells differ from Joos by 0.06–0.35 yr.
- *T3/T2 numerics (floats):* Joos, $d = 5$, $h(5) = 0.7712$: $\theta \in \{0.8, 0.95\}$ gives exactly two maxima and one minimum at the predicted $d + x^*$; $\theta \in \{0.5, 0.65, 0.7\} < h(d)$ gives one. For the $a_0 = 0$ IRF ($\tau = 100, 5$, $L = e^{-0.05}$): two maxima for $\theta \in (h(d), L)$, one outside. ρ is strictly increasing on a $[0, 500]$ grid (a double-precision run reported otherwise for $a_0 = 0$: float saturation of $e^{-x/5}$; at 60 digits it is strictly increasing).
- *Hand re-derivation:* Theorem 3 correct, with the incomplete step $u'(z_3+) < 0$ repaired and the exact count and necessity added; Theorem 2 correct as quasi-concavity, literally false as "at most one local maximum".

8 Audit corrections applied

The audit (`AUDIT.md`, §6) required twelve corrections to the producer's report (`data/RESULT.md`, which is the producer's *pre-audit* text and is superseded by this paper). They are applied here, in the numbering of §6:

1. **Headline of (c).** Corollary 4: "for every completely monotone, strictly decreasing IRF that is not a single exponential"; the single exponential is quasi-concave; "two maxima" defined as two local maxima separated by a strictly lower value (Section 1).
2. **Prior art.** Section 2: T2 is a special case of Ibragimov's strong-unimodality theorem; the existence half of (c) for integrable IRFs ($a_0 = 0$) follows from it; Ibragimov read from secondary sources only; the contribution is the explicit window, the certified instances and the framing; "to our knowledge", search not systematic.
3. **T2.** Theorem 2: non-decreasing then non-increasing (quasi-concave); the "at most one local maximum" wording is false for non-strict maxima (counterexample given).
4. **T3.** Theorem 3(iii)–(iv): $u'(z_3+) = -A(1 - h(z_3)) + B(1 - h(z_3 - d)) < 0$; exactly two maxima; the window is also necessary within the family; strict log-convexity by Cauchy–Schwarz on the Bernstein measure.
5. **I1, I2, S1.** Section 3.3: I1 and S1 are only weakly unimodal, I2 strictly; (c) holds under the strict definition through I2.
6. **Fit window.** Section 1, Theorem 1, control P5 and Section 4: the limit $a_0 W$, $u(10^6)$ and the $[0, 6000]$ yr scan describe the fitted formula beyond its 0–1000 yr range; certified peaks (≤ 287 yr) and times (≤ 500 yr) are inside.
7. **Seshadri wording.** Section 4: both readings of "0.015 %/year" lie outside $r > 1/\tau_m$; the 0.999 sum as a neutral note; not contacted.
8. **Table cell $r = 0.05$, $\tau_m = 20$.** Footnote † in Section 4: convention-dependent.
9. **Time origin and eq. (4).** Section 4: the producer's origin differs from Seshadri's, so comparisons with his figures are Tier [X]; the missing boundary term in his printed eq. (4) noted neutrally.
10. **Three-maximum counts.** Section 3.3 (Tier [X] paragraph): float-grid counts, not certified, not checked by the audit.

11. **Mechanism.** Section 3: with $a_0 > 0$ the second maximum is the end of the long low plateau; a model property only.
12. **Header and Lean status.** Disclosure and Section 6: independent audit `AUDIT.md` (ACCEPT_WITH_CORRECTIONS); Lean Tier A for discrete emission measures only, recompiled, axiom-audited and `leanchecker`-replayed by the auditor; dependency oleans reused, not rebuilt; statement lock owed.

9 Claim ledger

`ledger/build_ledger.py` files 15 claims: 2 at Tier [A] (`CK3-A-0001`, `-0002`), 3 at [L] (`CK3-L-0001..0003`), 8 at [C] (`CK3-C-0001..0008`, the last being the headline) and 2 at [X] (`CK3-X-0001`, `-0002`). Each carries a content-addressed evidence blob in `ledger/evidence/`; each Tier [A] claim also carries an `ai_audit` field. The gate field `audit` is left null: the audit was done by an AI agent, not a person, so the Elenchus gate keeps `LEDGER_UNAUDITED_TIER_A` open for each Tier [A] row. `ledger/run_gate.py` runs the unmodified gate; the verbatim output and exit code are in `ledger/gate_output.txt`.

10 Limitations and what remains unverified

- **Linear model only; no real-world claim.** No tonnes, climate effect, policy or "discovery". The Joos fit is a multi-model mean of one pulse experiment.
- **Classical mechanism.** Explicit, certified instances of a known phenomenon class (Ibragimov), not a new phenomenon; Ibragimov read from secondary sources only; the carbon-cycle literature search was not systematic.
- **AI audit only; human audit and statement lock owed.**
- **Unverified, from the audit.** (i) Global uniqueness of each peak in (b) is certified only within the initial 0.01-yr brackets; the global single-peak statement and the auditor's cross-check are float scans (Tier [X]), the auditor having checked 21 Joos scenarios and 6 of the 42 Seshadri-variant cells. (ii) The float maps in (c) (plateau table, 1500 random paths, three-maximum cases, $a_0 = 0$ and two-term checks) were not reproduced as certificates. (iii) The continuous-emission version of (a), T2 and T3 are paper proofs, checked by hand by the auditor. (iv) The preregistration (`prereg_K3.json`, sha256 `fc9571ec...`) hash matches, but its ordering relative to the work cannot be checked from git history. (v) The P4 arithmetic was checked only as the exact rational 5091/6250.
- **Deviations from the preregistration.** The Lean lemma set was extended with `cm_convexOn` and the limit statement (no change of claim); the second search budget for (c) was not needed; the exploratory scripts `explore3.py` and `explore_float.py` remain in `data/code/` (the producer's text said they were deleted; their hashes are in `result.json`); the instances were found with floats (`k3lib.u_float`) and then certified by `run_c.py`.
- **Outcome.** PARTIAL: (a) and (c) as preregistered (Lean for the discrete case only), (b) for a stated scenario family with a parameter inconsistency in the source.

11 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; the K3 outputs, the Lean files and this paper are uncommitted working-tree files at the time of writing (HEAD `d85a839`); nothing was pushed. Copies of the inputs are in `data/` (`RESULT.md`, `result.json`, `prereg_K3.json`, `AUDIT.md`, `code/`, `data/`, `logs/`, `lean/`, `audit_scripts/`). The producer ran on cores 4, 5 only. The bundler keeps only listed file suffixes, so the `.log`, `.sha256` and `.sage` files (the ledger hashes the originals) are also stored as byte-identical `.txt` copies in `data/`; compiled Lean objects are not shipped.

```
taskset -c 4,5 .venv/bin/python results/proposal4/2026-10-27/K3/code/run_b.py   # (b) enclosures
taskset -c 4,5 .venv/bin/python results/proposal4/2026-10-27/K3/code/run_c.py   # (c) certified instances
.venv/bin/python results/proposal4/2026-10-27/K3/code/check_two_term.py
```

```
.venv/bin/python results/proposal4/2026-10-27/K3/code/control_closed_form.py
python -I scripts/certified_numerics/lean_pinned.py compile formal_cert/Programmes/CarbonK3/Core.lean
python -I papers/p4_carbon_impulse_response/ledger/build_ledger.py
python -I papers/p4_carbon_impulse_response/ledger/run_gate.py
```

References

[1] F. Joos et al., Atmos. Chem. Phys. 13, 2793 (2013), doi:10.5194/acp-13-2793-2013, CC BY 3.0. Table 5 row IRFCO2, eqs. (7), (11) ([L]; the PDF is not redistributed).

[2] Seshadri, arXiv:1502.00805 (v3), later Global and Planetary Change 2017, doi:10.1016/j.gloplacha.2016.09.004. Eqs. (1), (2), (4), Sect. 3.2, Figs. 2 and 5 ([L]; the PDF is not redistributed).

[3] I. A. Ibragimov, *On the composition of unimodal distributions*, Teor. Veroyatnost. i Primenen. 1(2), 283–288 (1956); English translation in Theory Probab. Appl. 1. *Read only from secondary sources*, see [4].

[4] Encyclopedia of Mathematics, "Unimodal distribution"; Watanabe, Nagoya Math. J. (1991), on strong unimodality (cited as recorded in the audit; secondary sources for Ibragimov's theorem).

[5] Ciais et al., arXiv:2609.34226; only the printed numbers used in control P4 are quoted (licence not checked, nothing redistributed).